Reading in Data

The first step is to read in the Crowd Counting Consortium data. I downloaded the three available phases; 2017-2020, 2021-2024, and 2025's currently availiable data.

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(lubridate)
```

```
ccc_files <- list.files("data/raw/ccc", pattern = "\\.csv$", full.names = TRUE)
ccc_files
```

```
[1] "data/raw/ccc/ccc_compiled_20172020.csv"
[2] "data/raw/ccc/ccc_compiled_20212024.csv"
[3] "data/raw/ccc/ccc-phase3-public.csv"
```

I want to stack the three csv files into one table.

```r
ccc_raw <- ccc_files |>
  map(\(f) read_csv(f, col_types = cols(.default = col_character())) |>
        rename_with(tolower)) |>
   bind_rows()

names(ccc_raw)
```

```
  [1] "date"                    "locality"
  [3] "state"                   "location_detail"
  [5] "online"                  "type"
  [7] "macroevent"              "actors"
  [9] "claims"                  "valence"
 [11] "issues"                  "size_text"
 [13] "size_low"                "size_high"
 [15] "size_mean"               "size_cat"
 [17] "arrests"                 "arrests_any"
 [19] "injuries_crowd"          "injuries_crowd_any"
 [21] "injuries_police"         "injuries_police_any"
 [23] "property_damage"         "property_damage_any"
 [25] "chemical_agents"         "source_1"
 [27] "source_2"                "source_3"
 [29] "source_4"                "source_5"
 [31] "source_6"                "source_7"
 [33] "source_8"                "source_9"
 [35] "source_10"               "source_11"
 [37] "source_12"               "source_13"
 [39] "source_14"               "source_15"
 [41] "source_16"               "source_17"
 [43] "source_18"               "source_19"
 [45] "source_20"               "source_21"
 [47] "source_22"               "source_23"
 [49] "source_24"               "source_25"
 [51] "source_26"               "source_27"
 [53] "source_28"               "source_29"
 [55] "source_30"               "notes"
 [57] "lat"                     "lon"
 [59] "resolved_locality"       "resolved_county"
 [61] "resolved_state"          "fips_code"
 [63] "title"                   "organizations"
 [65] "participants"            "claims_summary"
 [67] "claims_verbatim"         "issue_tags_summary"
 [69] "issue_tags_verbatim"     "issue_tags"
```

```
 [71] "participant_measures"       "police_measures"
 [73] "participant_deaths"         "police_deaths"
 [75] "location"                   "event_type"
 [77] "notables"                   "targets"
 [79] "conf"                       "participant_injuries"
 [81] "police_injuries"            "participant_casualties_any"
 [83] "police_casualties_any"      "repeater"
 [85] "source1"                    "source2"
 [87] "source3"                    "source4"
 [89] "source5"                    "source6"
 [91] "source7"                    "source8"
 [93] "source9"                    "source10"
 [95] "source11"                   "source12"
 [97] "source13"                   "source14"
 [99] "source15"                   "source16"
[101] "source17"                   "source18"
[103] "source19"                   "source20"
[105] "source21"                   "source22"
[107] "source23"                   "source24"
[109] "source25"                   "source26"
[111] "source27"                   "source28"
[113] "source29"                   "source30"
```

```
nrow
```

```
function (x)
dim(x)[1L]
<bytecode: 0x1319da5f8>
<environment: namespace:base>
```

```
ccc_raw |>
  head()
```

```
# A tibble: 6 x 114
  date      locality state location_detail online type  macroevent actors claims
  <chr>     <chr>    <chr> <chr>           <chr>  <chr> <chr>      <chr>  <chr>
1 2017-01-~ Washing~ DC    Lafayette Squa~ 0      vigil <NA>       <NA>   for b~
2 2017-01-~ Mankato  MN    <NA>            0      vigil <NA>       Peace~ for p~
3 2017-01-~ Minneap~ MN    U.S. Bank Stad~ 0      prot~ <NA>       gener~ again~
4 2017-01-~ Little ~ RI    Town Green      0      vigil <NA>       Sakon~ for p~
5 2017-01-~ Oak Rid~ TN    Y-12 National ~ 0      vigil <NA>       Oak R~ for a~
```

```
6 2017-01-~ Washing~ DC    Lafayette Squa~ 0      vigil <NA>       <NA>   for b~
# i 105 more variables: valence <chr>, issues <chr>, size_text <chr>,
#   size_low <chr>, size_high <chr>, size_mean <chr>, size_cat <chr>,
#   arrests <chr>, arrests_any <chr>, injuries_crowd <chr>,
#   injuries_crowd_any <chr>, injuries_police <chr>, injuries_police_any <chr>,
#   property_damage <chr>, property_damage_any <chr>, chemical_agents <chr>,
#   source_1 <chr>, source_2 <chr>, source_3 <chr>, source_4 <chr>,
#   source_5 <chr>, source_6 <chr>, source_7 <chr>, source_8 <chr>, ...
```

I have run into a problem where the three phases all call the "protest inciting issue" different column names. The older phases use "issues," whereas the newer phases use "issues_tags."

```
ccc_raw |>
  summarise(
    issues       = sum(!is.na(issues)),
    issue_tags   = sum(!is.na(issue_tags)),
    tags_summary = sum(!is.na(issue_tags_summary)),
    total_rows   = n()
  )
```

```
# A tibble: 1 x 4
  issues issue_tags tags_summary total_rows
   <int>      <int>        <int>      <int>
1 131060     137452       198532     275151
```

```
ccc_raw <-
  ccc_raw |>
  mutate(topic = coalesce(issue_tags, issues, issue_tags_summary))

sum(is.na(ccc_raw$topic))
```

```
[1] 6639
```

```
ccc_raw |> count(topic, sort = TRUE) |> head(20)
```

```
# A tibble: 20 x 2
   topic                                                                  n
   <chr>                                                              <int>
 1 foreign affairs                                                    23772
 2 policing; racism                                                   11820
```

```
 3 military                                                                 11196
 4 policing;racism                                                           9671
 5 immigration                                                               7807
 6 education; guns                                                           7299
 7 labor                                                                     7008
 8 <NA>                                                                      6639
 9 democracy;presidency                                                      6163
10 presidency                                                                5757
11 environment                                                               5463
12 racism                                                                    4242
13 democracy;economy;education;environment;guns;healthcare;immigration;pr~  3564
14 religion;reproductive rights                                              3513
15 guns                                                                      3048
16 democracy                                                                 2962
17 healthcare;labor                                                          2948
18 foreign affairs;military                                                  2781
19 corruption;democracy;economy;education;environment;guns;healthcare;imm~  2507
20 lgbtqia                                                                   2466
```

```
sum(is.na(ccc_raw$topic))
```

```
[1] 6639
```

```
climate <- ccc_raw |>
  filter(str_detect(str_to_lower(topic), "environment|climate"))

nrow(climate)
```

```
[1] 23773
```

```
climate |> count(topic, sort = TRUE) |> head(15)
```

```
# A tibble: 15 x 2
   topic                                                                    n
   <chr>                                                                <int>
 1 environment                                                           5463
 2 democracy;economy;education;environment;guns;healthcare;immigration;pr~  3564
 3 corruption;democracy;economy;education;environment;guns;healthcare;imm~  2507
 4 democracy;economy;environment;healthcare;housing;immigration;labor and~   836
 5 development;environment;policing;racism                                741
```

```
 6 banking and finance;energy;environment                                    627
 7 energy;environment                                                        540
 8 energy;environment;indigenous peoples                                     361
 9 banking and finance;economy;energy;environment                            360
10 development;energy;environment                                           359
11 energy; environment                                                       352
12 civil rights; disability rights; economy; environment; immigration; la~   332
13 development; economy; energy; environment; indigenous peoples; labor; ~   293
14 environment;indigenous peoples                                            225
15 civil rights;economy;environment;guns;immigration;labor and workers ri~   219
```

I want to interrogate the distribution of untagged events to decide what I will do with the missing data.

```
ccc_raw |>
  filter(is.na(topic)) |>
  mutate(year = str_sub(date, 1, 4)) |>
  count(year)
```

```
# A tibble: 10 x 2
   year      n
   <chr> <int>
 1 2017   1081
 2 2018   1530
 3 2019    567
 4 2020    409
 5 2021    594
 6 2022    616
 7 2023    545
 8 2024    616
 9 2025    442
10 2026    239
```

```
mean(is.na(ccc_raw$topic))
```

```
[1] 0.02412857
```

I have decided that I will exclude events with no issue tags (about 2.4% of events, spread across all years)

Lets now count how many climate and environmental protests happened in each month. This will allow me to find the protest waves (a month in the top 10% of monthly counts), which will tell me what wikipedia data I need to download.

```
climate <-
  climate |>
  mutate(
    date  = as_date(parse_date_time(date, orders = c("ymd", "mdy"))),
    month = floor_date(date, "month")
  )

nrow(climate)
```

```
[1] 23773
```

```
range(climate$date, na.rm = TRUE)
```

```
[1] "2017-01-01" "2026-05-31"
```

```
sum(is.na(climate$date))
```

```
[1] 0
```

```
climate <- climate |>
  mutate(
    date  = as_date(parse_date_time(date, orders = c("ymd", "mdy"))),
    month = floor_date(date, "month")
  )

nrow(climate)
```

```
[1] 23773
```

```
range(climate$date, na.rm = TRUE)
```

```
[1] "2017-01-01" "2026-05-31"
```

```r
sum(is.na(climate$date))
```

```
[1] 0
```

```r
monthly <-
  climate |>
  count(month, name = "protests")
monthly |> arrange(month)
```

```
# A tibble: 113 x 2
   month      protests
   <date>        <int>
 1 2017-01-01       29
 2 2017-02-01       58
 3 2017-03-01       49
 4 2017-04-01      123
 5 2017-05-01       22
 6 2017-06-01       38
 7 2017-07-01       28
 8 2017-08-01       20
 9 2017-09-01       19
10 2017-10-01       27
# i 103 more rows
```

```r
head(10)
```

```
[1] 10
```

```r
exists("climate")
```

```
[1] TRUE
```